

FINANCIAL INNOVATION LAB

INTRODUCTION ANTI MONEY LAUNDERING

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Perpetrators of fraud

2,410 cases reported

Gender

69% Male

31% Female



Position/title

41% Employee

37% Manager

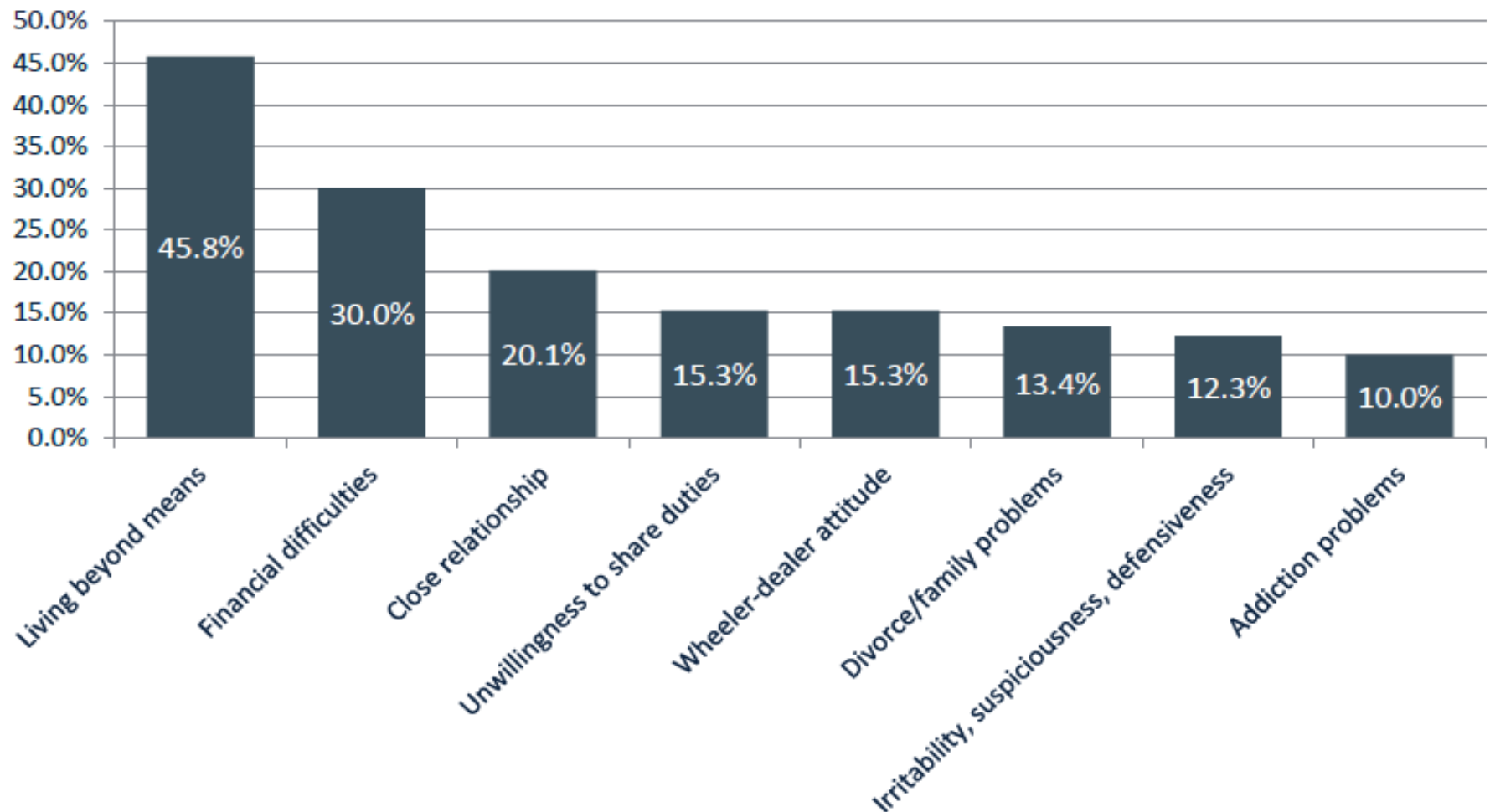
19% Owner/officer

3% Other

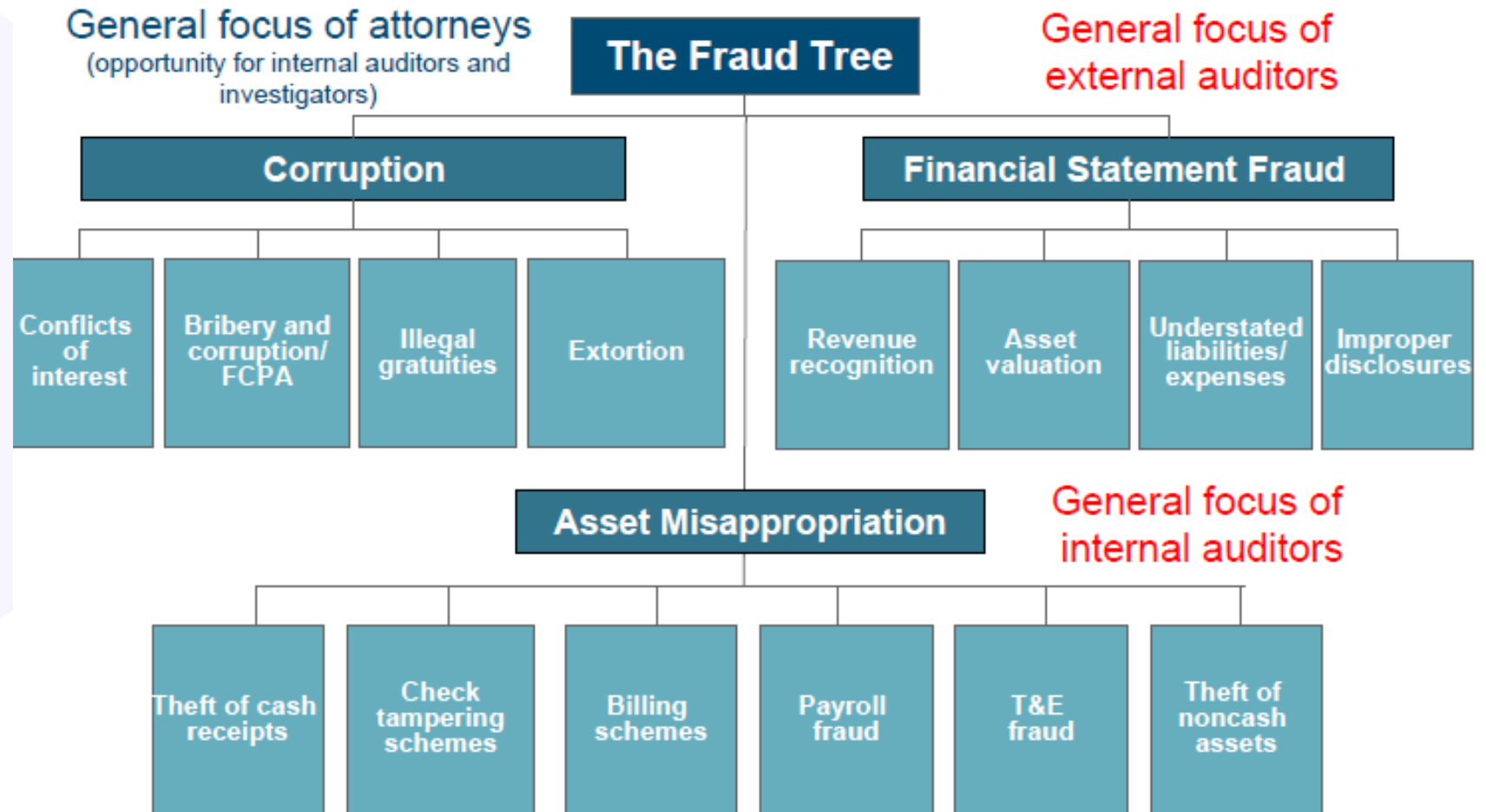
2016 ACEE Report to the Nations on Occupational Fraud and Abuse

Occupational fraud – perpetrators

Behavioral Red Flags of Perpetrators



Planning: Build a Profile of Potential Frauds



Grote financiële fraudes

- 1. Enron** (gasleverancier) heeft opzettelijk een verkeerde voorstelling gegeven van winsten, kasstromen en ondergewaardeerde verplichtingen met behulp van creatieve, maar twijfelachtige boekhoudmethoden. Een verkapte lening in 1999 waarbij de opbrengst van de verkoop van obligaties als kasstroom uit bedrijfsactiviteiten werd verantwoord. Deze overschatte operationele cashflow met \$ 700 miljoen. Door gebruik te maken van market to market accounting, nam Enron een zeer aanzienlijk bedrag aan toekomstige inkomsten op als lopende inkomsten. Hierdoor kon een bepaalde business unit een kwartaalwinst van \$ 40 miljoen rapporteren, terwijl deze unit in feite met verlies werkte. Een andere leningentransactie werd met \$ 4,85 miljard ondergewaardeerd. Arthur Anderson failliet.
- 2. Ahold** Financiële verantwoording over Amerikaanse activiteiten veel te rooskleurig (2001). In VS schikking van 937 miljoen euro getroffen (2006). Individuele bestuurders aangeklaagd.

Detecteren van money laundering (wettelijke plicht voor banken)

Informatiebronnen

Dataset met bank transacties

Digitale controle

AI-modellen

Red Flags

Verdachte transacties

Check Red Flag

Handmatige controle

Financiële fraude?

Imbalance of dataset

Data: Features in bank transactions

Unsupervised

- Time and Date: Essential for tracking transaction chronology.
- Sender and Receiver Account Details: Helps uncover behavioral patterns and complex banking connections.
- Amount: Indicates transaction values to identify suspicious activities.
- Payment Type: Includes various methods like credit card, debit card, cash, ACH transfers, cross-border, and cheque.
- Sender and Receiver Bank Location: Pinpoints high-risk regions including Mexico, Turkey, Morocco, and the UAE.
- Payment and Receiver Currency: Align with location features, adding complexity when mismatched.

Supervised

- 'Is Suspicious' Feature: Binary indicator differentiating normal from suspicious transactions.
- Type: Classifies typologies, offering deeper insights.

Red flags of money laundering can be indicated by

- a. Round Amount payments
- b. Payments from high-risk countries (<https://www.fatf-gafi.org/en/countries/black-and-grey-lists.html>) to tax havens (<https://fsi.taxjustice.no/fsi/2022/world/score/top>)
- c. Smurfing
- d. Nesting
- e. Shell company characteristics using address and name data
- f. Non-adherence to FATF Recommendation (<https://www.fatf-gafi.org/content/dam/recommandations/pdf/FATF%20Recommendations%202012.pdf.coredownload.inline.pdf>)

Suspicious transactions (1)

Structuring	Splitting a large amount into many smaller transactions to avoid reporting thresholds.
Cash_Withdrawal	Rapid or repeated large cash withdrawals, often following inbound transfers.
Deposit-Send	Quick deposit followed by immediate outward transfers, with little or no balance seasoning.
Smurfing	Many small deposits from multiple sources funnelled into one account to evade detection.
Layered_Fan_In	Funds converge through multiple intermediaries into a single node (layered aggregation).
Layered_Fan_Out	Funds disperse from one source through several layers to obscure the transaction trail.

Suspicious transactions (2)

Stacked Bipartite	Two disjoint groups transact densely across the partition in staged layers.
Behavioural_Change_1	Sudden increase in volume, frequency, or counterparties inconsistent with past behaviour.
Bipartite	Heavy traffic between two clusters with minimal internal flows, indicating possible conduit structures.
Cycle	Funds exit and return to their origin through a closed loop (round-tripping).
Fan_In	Many unrelated payers funnel funds into a single account without clear legitimate rationale.
Gather-Scatter	Funds gathered into a hub and quickly dispersed to many endpoints.

Suspicious transactions (3)

Behavioural_Change_2 Alternative form of abrupt behavioural deviation relative to historical patterns.

Scatter-Gather Funds distributed to many receivers and later recombined at one or few endpoints.

Single_large An isolated very large transfer anomalous for the account's history.

Fan_Out One account sends to many recipients at unusual velocity or amounts.

Over-Invoicing Transfers justified by inflated invoices relative to plausible goods or services.

Basic AI Methods

Unsupervised

Supervised

XGBoost

Logistic Regression

Random Forest

Decision Trees

K-Nearest Neighbors (KNN)

Artificial Neural Networks (ANN)

Performance measures of algorithms

Imbalance of dataset: many transactions, few money laundering transactions.

- 1) Accuracy = percentage correct classifications
$$= (TP + TN) / Total$$
- 2) Precision = percentage correct positive classifications
$$= TP / (TP + FP)$$
- 3) Recall = ratio of 'correct positives classifications' and 'all positives'
$$= TP / (TP + FN)$$
- 4) F1 = (precision * recall) / (precision + recall)
- 5) F2 = recall is more important than precision